PRE-CLASS PREP

Electrolytes & Fluid Balance

Know Before You Go

Read this before class — it will make the lecture click faster.

Big picture: Electrolytes keep nerves firing, hearts beating, and muscles contracting. Small shifts outside the normal range can be life-threatening. Today covers sodium, potassium, calcium, magnesium, and fluid volume — know your normals, know your signs, know your interventions.

POTASSIUM — MOST DANGEROUS IMBALANCE

Hypokalemia <3.5 — LOW
Causes: loop/thiazide diuretics, vomiting, diarrhea, NG suction, excessive alcohol, chronic kidney disease
S&S: muscle weakness, respiratory paralysis, cardiac arrhythmias, ↓reflexes, **flat/inverted T wave**
Tx: cardiac monitor; oral K^+ foods; IV K^+ if critically low — **NEVER IV push; MAX 20 mEq/hr**

Hyperkalemia >5.0 — HIGH
Causes: renal failure, K^+-sparing diuretics, NSAIDs, sepsis, excessive intake
S&S: muscle weakness, chest pain, cardiac dysrhythmias, **peaked T waves**, cardiac arrest
Tx: cardiac monitor; calcium gluconate (stabilizes heart); IV insulin (shifts K^+ into cells); kayexalate; dialysis if renal failure

SODIUM

Hyponatremia <135 — LOW
Causes: excess water, SIADH, heart/kidney/liver failure, diuretics, vomiting
S&S: nausea, confusion, headache, restlessness, seizures (brain cells swell)
Tx: fluid restriction; antiemetics/antidiarrheals; avoid gum and hidden fluid sources; **correct slowly**

Hypernatremia >145 — HIGH
Causes: DI, dehydration, fever, vomiting, gastroenteritis, impaired thirst, CKD
S&S: confusion, thirst, ↓urine, ↓LOC, seizures (brain cells shrink)
Tx: oral fluids if mild; hypotonic IV fluids; **decrease sodium slowly** — rapid correction causes cerebral edema

CALCIUM

Hypocalcemia <8.4 — LOW
Causes: ↓Vit D, hypoparathyroidism, renal disease, hormonal changes
S&S: numbness/tingling, muscle cramps, tetany, seizures, **+ Chvostek's & Trousseau's**, chest pain, dysrhythmias
Tx: seizure precautions; oral Ca^{2+} + Vit D; IV calcium gluconate if severe; correct hypomagnesemia concurrently

Hypercalcemia >10.5 — HIGH
Causes: hyperparathyroidism, cancer, prolonged bed rest, Vit D toxicity
S&S: GI symptoms first (N/V, abd pain), confusion, thirst, bone pain, muscle weakness
Tx: IV NS + loop diuretics; encourage ambulation; fall precautions; seizure precautions if severe

MAGNESIUM

Hypomagnesemia <1.5 — LOW
Causes: loop/thiazide diuretics, PPIs, malnutrition, diarrhea, alcoholism, decreased intestinal absorption
S&S: mirrors hypocalcemia — N/V, fatigue, tetany, dysrhythmias, + Chvostek's & Trousseau's
Tx: oral Mg^{2+}; IV magnesium sulfate if severe — **high-alert medication, double-check dose**; monitor respiratory rate; correct hypokalemia/hypocalcemia concurrently

Hypermagnesemia >2.5 — HIGH
Causes: renal failure, antacids/laxatives with Mg^{2+}, trauma, hypothyroidism
S&S: ↓DTRs (early warning), hypotension, flushing, headache, blurred vision, respiratory depression, cardiac arrest
Tx: IV calcium gluconate (antidote); loop diuretics; IV fluids; cardiac monitor; dialysis if severe

- **Key link:** low magnesium → cannot correct low calcium or low potassium until magnesium is replaced first
- **Loss of DTRs = early magnesium toxicity warning** — notify provider immediately before giving more magnesium

FLUID VOLUME

Deficit (Hypovolemia)
Causes: vomiting, diarrhea, bleeding, burns, sweating, polyuria, fever, trauma
S&S: ↑HR, ↓BP, dry membranes, poor skin turgor, ↑urine specific gravity, ↑BUN/Hgb/Hct
Tx: treat cause; oral rehydration if mild; IV fluids if moderate/severe; hypovolemic shock at >20% fluid loss

Excess (Hypervolemia)
Causes: heart failure, kidney/liver failure, nephrotic syndrome, excessive IV fluids
S&S: ↑BP, bounding pulse, crackles, JVD, pitting edema, weight gain, ↓BUN/Hgb/Hct
Tx: treat cause; diuretics; fluid and sodium restriction; daily weights; dialysis if renal failure